



Token Migration Submission Infrastructure

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Moving to tokens in Submission Infrastructure

- Motivations
 - Towards **industry** standards
 - Identity mapping (old) vs **capabilities** (new)
 - Globus Toolkit **retirement**
 - Practical example: *multiple tokens with different capabilities instead of a single powerful pilot (GSI) proxy*
- Timeline
 - WLCG/OSG dictating the pace (most recent [timeline](#))
 - Important dates for Submission Infrastructure
 - **May 1st**: OSG 3.6 release will remove any Globus toolkit dependency
 - **November 2022**: HTCondor GSI End Of Life

Transitions for Submission Infrastructure:

1. Sub. Inf. internal components communication
2. Factory to Sites communication



Global Pool internal component communication

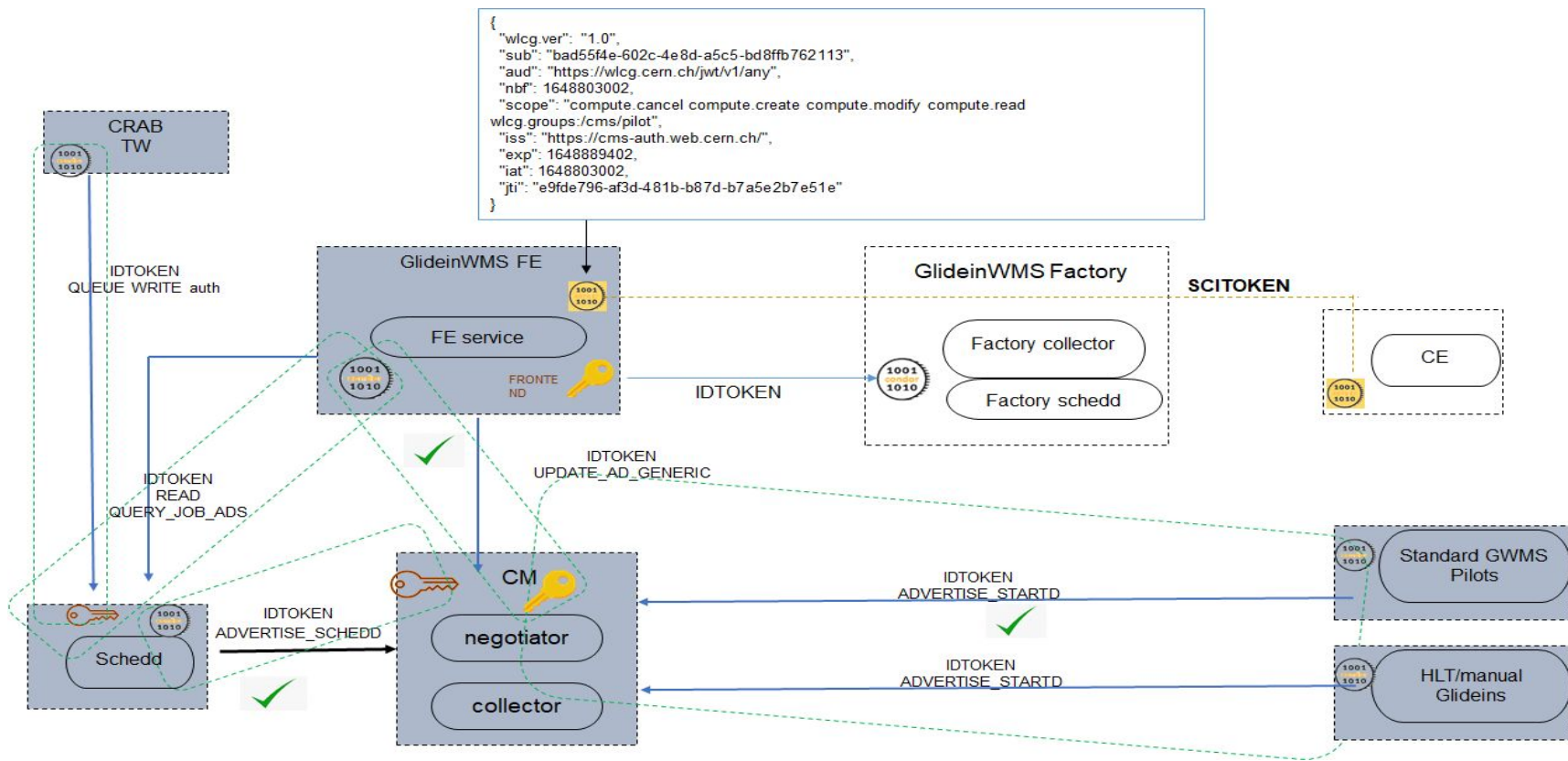


HTCondor and Tokens

- With GSI End of Life (November) **HTCondor** will **only** use **IDTOKENS** for authentication/authorization. E.g.:
 - CRAB/WMAgent to schedulers for job submission
 - Scheduler to central manager
 - Worker node to central manager
 - GlideinWMS frontend to central manager/schedulers
- IDTOKENS created using **symmetric keys**
 - Only token issuer can validate the token
 - **Signing key can be shared** between components
 - GWMS frontend generates IDTOKEN with shared signing key
 - Tokens travel all the way to the worker nodes
 - Worker nodes connects to central manager that verifies token since it has the same signing key as the frontend



Token authentication Implementation





HTCondor and Tokens

- The whole infrastructure is **already using tokens** for authentication!
 - Tokens enabled in ITB first as usual
 - Incremental deployment
 - GSI is still enabled and used as fallback on some cases
 - E.g.: tail of 6k jobs (out of 170k) use GSI fallback
- Also worked with teams using manually launched pilots to enable WN to CM authentication
 - HTL, BSC, Lancium, Marconi100, etc

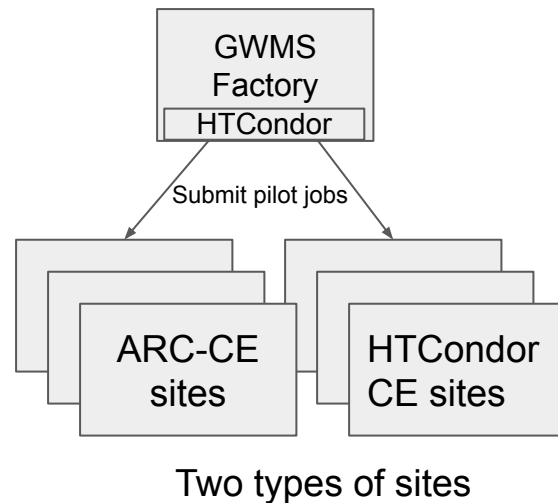


Factory to Sites communication



Ways to submit jobs to the Grid

- Factory submits pilots to sites that use two CE technologies using HTCondor
- ARC-CEs submission:
 - Through LDAP interface and GSI proxies (historical)
 - **New REST interface** available
 - Supports **tokens** but they **can't be used yet**
 - Plain X509 proxies can be used (no GSI)
- HTCondor submission
 - Through GSI proxies (historical)
 - **New** versions support **SCITOKENS**





Transition consequences

- When OSG 3.6 “GSI-less” will land (May 1st):
 - OSG sites will **only** accept pilot submission via **tokens**
 - UCSD factory will stop supporting GSI (CERN factory will still support it)
 - Need to “**partition pilot submission**”
 - Sites still using GSI configured in the CERN factory
 - Sites that uses **tokens (HTCondorCEs)** or the **ARC REST interface** in the UCSD factory
- HTCondor GSI End Of Life (November 2022):
 - Directly impact HTCondorCEs
 - Indirectly impact ARC-CEs since factory uses condor to submit pilot jobs
 - **Deadline for stopping usage of GSI proxies**



Actions taken

- Round of **tests in ITB**
 - Multiple HTCondor and GWMS versions tested
 - Feedback to devs
- 3.7.6 version of GWMS with **full token support on Jan 26th**
 - 3.7.7 with improvements in the pipeline (e.g.: one token per site)
- Defined strategy for deployment of token (see shared [Document](#))
 - Minimize disruption (profiting from GSI fallback mechanism)
 - Work systematically with sites and only start using tokens when fully validated
- Necessary **configurations** and software **shipped in OSG 3.5 on the 10th March**
 - Thanks to the OSG software team for the collaboration!
- Opened **GGUS tickets to sites** on 15th March



Current status

- Three categories of tickets

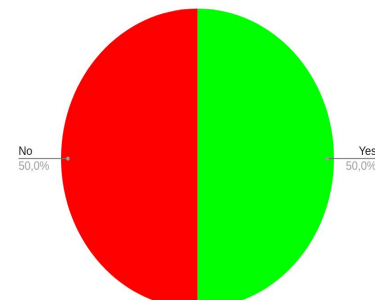
- **OSG** sites with CEs **not accepting tokens** and other sites with recent enough versions
- Sites with **old versions of HTCondor**
 - What were the plans for upgrade?
- Run **tests on ARC CE** sites to check if REST interface is working
 - Waiting for token support in HTCondor to open tickets (end of April).

- Summarized in [spreadsheet](#)

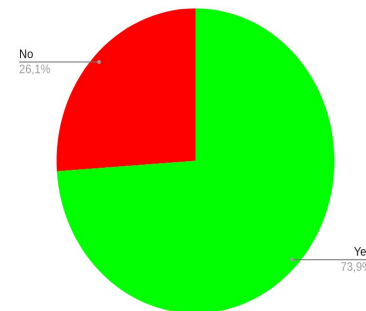
- OSG T1 and T2s (9 US + 2 BR)
 - 6 out of 11 completed transition
 - All with at least one CE with working ping
- OSG T3 and OSG opportunistic sites in a worse shape
 - Contacted all of them but slow progress

May deadline

HTCondorCE accepts token (basic test)



ARC REST interface works (basic test)





Conclusions

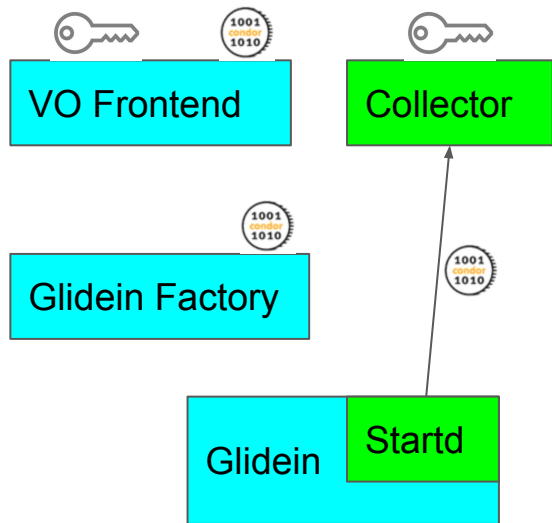
- CMS Submission Infrastructure **internal components already switched to IDTOKENS**
- **Working with sites** to guarantee a seamless transition **for Factory<->CE communication**
 - Some of OSG T3 might not meet the May 1st deadline but we will keep using them through the CERN factory if possible
- Site Support Team for **long term support**
 - Some sites stopped accepting tokens after we enabled them (GSI fallback still works)



Backup slides



Replacing proxy with IDTOKEN for CM auth/authz



- **Same signing key** placed on both **Frontend** and **Collector**
- Frontend generates an IDTOKEN
- **IDTOKEN is transferred** to the factory, the CE, Batch System, **WN** (as pilot proxy before)
- **Startd can be authenticated and authorized** by the collector

Other HTCondor related auth/authz that have been successfully tested



Replacing proxy with SCITOKENS for CE auth/authz

Proxy approach:

```
voms-proxy-init -voms cms:/cms/Role=pilot -valid 72:0
```

```
voms-proxy-init -voms cms:/cms/ local/Role=pilot  
-valid 72:0
```

- Two proxy generated and renewed through custom cronjob
 - Special proxy to be used to match “extra pledge” resources to local users
- Proxy copied to the Glidein factory and used for submission to CEs
 - Same proxy used for WN (startd) to Condor central manager communication (collector).
- Mapping on the CE (through Argus or lcmsaps):

```
"/cms/Role=pilot/Capability=NULL" cmspilot  
"/cms/local/Role=pilot/Capability=NULL"  
cmslocal
```

Token approach :

```
curl -s --user IAM_CLIENT_ID:IAM_CLIENT_SECRET -d scope="compute.read compute.read  
compute.modify compute.create compute.cancel wlcg.groups:/cms/pilot"  
https://cms-auth.web.cern.ch/token
```

```
curl -s --user IAM_CLIENT_ID:IAM_CLIENT_SECRET -d scope="compute.read compute.read  
compute.modify compute.create compute.cancel wlcg.groups:/cms/ local/pilot"  
https://cms-auth.web.cern.ch/token
```

- Tokens generated with curl
 - clients registered in IAM
- Token copied to the Glidein factory and used for submission co CEs
 - A different token has to be used for WN to Condor central manager communication. More later
- Mapping on the CE (through condor_mapfile):

```
SCITOKENS  
/^https:\\/\\/wlcg\\.cloud\\.cnaf\\.inf\\.it\\/,6e0ebc78-7015-45cc-88e0-7  
bb4683b02a6/ pilcms001
```